Iron Kaggle

Group:
Group 3

Members:
Rima Krivickiene; Philippe Rallier

Iron Kaggle

Machine Learning Bootcamp Final Project

01 Executive Summary

02 Methods (Preprocessing)

03 Methods (Models) — Part 1

04 Methods (Models) — Part 2

05 Selected Model

Summary

Best model is Random Forest.

Best model performance

0.9894
R2 (Train)

0.9416
R2 (Test)

Real-life prediction (our forecast)

R2 = 0.93

Alternatives tested (quick recap)

Linear Regression	R2 Test:	0.8523
KNN Regression	R2 Test:	0.8830
Decision Tree	R2 Test:	0.8561

Methods – Preprocessing

Dataset overview

- 640,840 rows
- 10 columns
- Missing values: none detected

Cleaning & feature engineering

- Removed `id` (no predictive value)
- Date handling: extracted year / month / day from `date`
- Categorical handling: `get_dummies` for `state_holiday`

Methods – Models Part 1

Models tested:

- Linear Regression
- KNN Regression
- Decision Tree
- Random Forest

Why these models:

- Progression from simple → more complex to reduce overfitting risk
- Linear Regression used as baseline
- Tree-based models tested to capture more complex patterns

Methods (Models — Part 2)

[R2 scores]

Selected Model

Winner model is Random Forest.

R2 (Train)	R2 (Test)	Real-life prediction
0.9894	0.9416	0.93

Why it won

- Highest R2 test score (0.9416)
- Good train/test balance overall
- Slight overfitting observed (0.9894 vs 0.9416), acceptable

Takeaways

Recap / Conclusions

- Predicted daily sales using Random Forest
- Achieved R2 (Test) = 0.9416

Challenges

- Date format handling
- Overfitting risk
- Data leakage prevention

Key learnings

- Importance of preprocessing
- Value of systematic model comparison
- Understanding bias–variance tradeoff

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Thank You